Lucas Teixeira Correia

Computer Science Student | Data, ML and AI Projects

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SUMMARY

Computer Science undergraduate at PUC-GO with hands-on experience in Python, SQL, exploratory data analysis, model evaluation and applied AI prototyping. Developed academic and personal projects involving optimization, Gaussian Process Regression (GPR), Efficient Global Optimization (EGO), API integration and structured handling of business data. Strong interest in data-driven problem solving, experimentation and building reliable AI solutions grounded in real use cases.

EDUCATION

Pontifical Catholic University of Goiás (PUC-GO) - B.Sc. in Computer Science

Goiânia, Brazil | Jan 2023 - Jul 2027 (expected) | GPA: 3.76/4.0

- Relevant coursework and practice in programming, algorithms, databases, software engineering, statistics and machine learning.

EXPERIENCE

Teaching Assistant - Algorithms | PUC-GO

Goiânia, Brazil | Aug 2023 - Dec 2023

- Supported students with programming exercises in C/C++ and core algorithm and data-structure concepts.
- Helped explain problem-solving strategies, code debugging and computational thinking in lab and class support activities.

Teaching Assistant - Calculus I | PUC-GO

Goiânia, Brazil | Jan 2024 - Jul 2024

- Assisted students in tutoring sessions focused on mathematical reasoning, functions, limits and derivatives.
- Strengthened quantitative foundations relevant to statistics, optimization and analytical modeling.

SELECTED PROJECTS

Optimization of Shallow Foundations with Metaheuristics and ML | Research Project - PUC-GO

May 2025 - Aug 2026

- Developing software for shallow-foundation optimization with metaheuristics, structural constraints and computational experimentation.
- Designed and evaluated experiments with GPR and EGO; achieved about 95% predictive accuracy in one objective-function modeling stage.
- Documented requirements, assumptions, constraints and acceptance criteria with advisor alignment.

Enterprise WhatsApp AI Assistant | Personal Project

2026 - Present

- Developing a business-oriented WhatsApp assistant in Python/FastAPI with OpenAI API integration.
- Separated company data contexts and structured retrieval/prompt flows for more reliable responses, with exposure to fine-tuning-related workflows and API evaluation.

Lung Cancer Dataset Review | Academic Exercise

2025 - 2026

- Reviewed a Kaggle lung-cancer dataset in a course exercise, performed initial exploratory analysis and ran baseline Logistic Regression tests to practice preprocessing and evaluation.
- Observed limitations in dataset quality and treated the activity as an introductory validation exercise rather than a production-grade modeling project.

Introductory ML Practice Studies | Personal Learning Projects

2025 - 2026

- Built small learning projects with KNN, multiple linear regression and model-comparison workflows to strengthen preprocessing, evaluation and interpretation skills.
- Used these exercises to reinforce core ML concepts and improve practical familiarity with scikit-learn-based experimentation.

ACHIEVEMENTS

- 2nd place - III Hacka PUC-GO Challenge (2023).

- 3rd place - JCPOLI Programming Marathon, PUC-GO (2023).
- 2nd place - Metaheuristics Optimization Challenge, PUC-GO (2025): developed Bin Packing solvers using GRASP and Simulated Annealing and explored hybridization with ML ideas.

TECHNICAL SKILLS

Languages: Python, SQL, Java, JavaScript, R, C++
Data and ML: Pandas, scikit-learn, Matplotlib, Plotly, statistics, feature engineering, model evaluation, exploratory data analysis
AI and LLMs: OpenAI API integration, prompt and context design, applied AI prototyping, fine-tuning workflow exposure, RAG concepts
Data Engineering and Tools: Relational databases, ETL concepts, data cleaning, Git, GitHub, Docker, Streamlit, Power BI, Jira
Optimization: Metaheuristics, GPR, EGO, experimental design
Languages: Portuguese (native), English (advanced/C1), German (beginner/A1)